

PARAMETRIC STUFF

For a curve C , introduce new parameter t s.t. $x=f(t)$ and $y=g(t)$ called **PARAMETRIC EQUATIONS**. As t varies $(x,y)=(f(t),g(t))$ trace curve C called **PARAMETRIC CURVE**

1) a set of parametric eqs can describe SAME CURVE but not describe same **PARAMETRIC CURVE**!

2) A **CURVE** is a set of points in which points are traced in a particular way as parameter changes within $b_0 \leq t \leq t_{max}$

Tangent line at (x_0, y_0) to $\vec{r}(t)$ is $\vec{v}(t)$

1) $y - y_0 = F'(x_0, y_0)(x - x_0)$
2) $F(x) = \frac{dy}{dx}$ when $\frac{dy}{dt} \neq 0$ and $\frac{dx}{dt} \neq 0$ and $\frac{dy}{dx} = 0 \Rightarrow$ **HORIZONTAL TANGENT**
if $\frac{dx}{dt} = 0$ and $\frac{dy}{dt} \neq 0 \Rightarrow$ **VERTICAL TANGENT**

3) evaluate $F(x)$ @ $x = x_0$, $y = y_0$

concave up or down:

CONCAVE UP: $\frac{d^2y}{dx^2} > 0$ CONCAVE DOWN: $\frac{d^2y}{dx^2} < 0$

AREA = $\int_0^b g(u) f'(u) du$ at where $a = f(a)$

ARC LENGTH L For parametric curves: $L = \int_a^b \sqrt{f'(t)^2 + g'(t)^2} dt$

$L = \lim_{n \rightarrow \infty} \sum_{i=1}^n \sqrt{1 + (\frac{\Delta y_i}{\Delta x_i})^2} \Delta x_i = \int_a^b \sqrt{1 + (\frac{dy}{dx})^2} dx = \int_a^b \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt$

SURFACE AREA

rotation about the x -axis rotation about the y -axis
 $S = \int_a^b 2\pi y ds = \int_a^b 2\pi y \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt$ $S = \int_a^b 2\pi x ds = \int_a^b 2\pi x \sqrt{(\frac{dx}{dt})^2 + (\frac{dy}{dt})^2} dt$

POLAR STUFF

Polar $\rightarrow$ Cart: $x = r \cos \theta$ Cart $\rightarrow$ Polar: $r = \sqrt{x^2 + y^2}$

$(r, \theta) = (-r, \theta + \pi)$ $y = r \sin \theta$ $\tan \theta = \frac{y}{x}$

tangents to $r = r(\theta)$: $x = r(\theta) \cos \theta$ $y = r(\theta) \sin \theta$ $\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$

AREAS/LENGTHS

R bounded by: 1) $r = f(\theta)$ 2) rays $\theta = a$ $\theta = b$ $0 < b - a \leq 2\pi$

$\Delta A_i \approx \frac{1}{2} [f(\theta_i^*)]^2 \Delta \theta$ $A \approx \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta$

$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{2} [f(\theta_i^*)]^2 \Delta \theta = \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta$

GENERAL: $r = f(\theta)$ $a \leq \theta \leq b$ $r = f(\theta)$ $\theta = a$ $\theta = b$ $A = \frac{1}{2} \int_a^b [f(\theta)]^2 d\theta$

$x = r \cos \theta$ $y = r \sin \theta$ $r = f(\theta)$ $\frac{dx}{d\theta} = -r \sin \theta + f'(\theta) \cos \theta$ $\frac{dy}{d\theta} = r \cos \theta + f'(\theta) \sin \theta$ $L = \int_a^b \sqrt{r^2 + (f'(\theta))^2} d\theta$

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CLAUDE'S THEOREM: Suppose f defined on disk D that contains $(0,0)$. If f is harmonic on D , then $f(x,y) = f_0(x,y) + f_1(x,y)$ where f_0 is harmonic and f_1 is linear.

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VECTOR FUNCTIONS

DOMAIN: set of real ts position vectors $\vec{r}(t)$ of points P w/ coordinates

$\vec{r}(t) = (f(t), g(t), h(t))$ $t \in I$ RANGE: set of vectors $\vec{r}(t)$ of points P trace **SPACE CURVE**

$\lim_{t \rightarrow a} \vec{r}(t) = \lim_{t \rightarrow a} (f(t), g(t), h(t)) = (\lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t))$ $\vec{r}(t)$ continuous at $t=a$ if $\lim_{t \rightarrow a} \vec{r}(t) = \vec{r}(a)$

$\frac{d\vec{r}}{dt} = \left(\lim_{h \rightarrow 0} \frac{\vec{r}(t+h) - \vec{r}(t)}{h} \right) = \left(\lim_{h \rightarrow 0} \frac{f(t+h) - f(t)}{h}, \lim_{h \rightarrow 0} \frac{g(t+h) - g(t)}{h}, \lim_{h \rightarrow 0} \frac{h(t+h) - h(t)}{h} \right) = \left(f'(t), g'(t), h'(t) \right)$

$\vec{r}(t)$ is TANGENT VECTOR to curve defined by $\vec{r}(t)$ (also a vector func) $\vec{v}(t) = \frac{d\vec{r}}{dt}(t) = (f'(t), g'(t), h'(t))$

$\frac{d}{dt} (\vec{u}(t) + \vec{v}(t)) = \frac{d\vec{u}}{dt}(t) + \frac{d\vec{v}}{dt}(t) = \vec{u}'(t) + \vec{v}'(t)$ $\frac{d}{dt} (\vec{u}(t) \cdot \vec{v}(t)) = \vec{u}'(t) \cdot \vec{v}(t) + \vec{u}(t) \cdot \vec{v}'(t)$ $\frac{d}{dt} (\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$

$\frac{d}{dt} (c \vec{u}(t)) = c \frac{d\vec{u}}{dt}(t) = c \vec{u}'(t)$ $\frac{d}{dt} (f(t) \vec{u}(t)) = f'(t) \vec{u}(t) + f(t) \vec{u}'(t)$ $\frac{d}{dt} (\vec{u}(t) \times \vec{v}(t)) = \vec{u}'(t) \times \vec{v}(t) + \vec{u}(t) \times \vec{v}'(t)$

$\int \vec{r}(t) dt = \left[\int f(t) dt \right] \vec{i} + \left[\int g(t) dt \right] \vec{j} + \left[\int h(t) dt \right] \vec{k}$

PARTIAL DERIVATIVES

func of 2 variables: assigns (x,y) in set D to unique $\mathbb{R}$ denoted by $f(x,y)$ **DOMAIN**: set of D

$z = f(x,y)$ (x,y) : INDEPENDENT VARIABLE z : DEPENDENT VARIABLE **GRAPH** of f : $\{(x,y,z) \in \mathbb{R}^3 : z = f(x,y)\}$ $(x,y) \in D$

LEVEL CURVES: curves defined by $f(x,y) = k$, where k is a constant (curves of const height k plotted in xy)

func of 3 variables: assigns (x,y,z) in $D \subset \mathbb{R}^3$ to unique $\mathbb{R}$ denoted by $f(x,y,z)$

LEVEL SURFACES: surfaces w/ equations $f(x,y,z) = k$ k is a constant

$\lim_{(x,y,z) \rightarrow (a,b,c)} f(x,y,z) = L$ if for every number $\epsilon > 0$, there is a corresponding number $\delta > 0$ s.t. if $(x,y,z) \in D$ and $\sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} < \delta$ then $|f(x,y,z) - L| < \epsilon$

$\lim_{x \rightarrow a, y \rightarrow b} f(x,y) = L$ or $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$

generally: if C, B, C_2 are diff. paths and if $f(x,y) \rightarrow L$ as $(x,y) \rightarrow (a,b)$ along path C_1 and $f(x,y) \rightarrow L_2$ as $(x,y) \rightarrow (a,b)$ along path C_2 where $L_1 \neq L_2$ then $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ DOES NOT EXIST.

CONTINUITY

f is continuous at (a,b) if $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$ $\Rightarrow$ f is continuous at (a,b) if f is continuous at every point on D w/ small amt, $f(x,y)$ changes small amt.

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CONTINUITY OF COMPOSITE FUNCTIONS $h = g \circ f(x,y)$ or $h = g \circ f$

if f is continuous on domain D and g is continuous at $f(a,b)$ then h is continuous on the domain D of f

if g is continuous at $f(a,b)$ and f is continuous on the domain D of f

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DOUBLE INTEGRAL IN POLAR COORDINATES

If f is continuous on a polar rect $R: \{(r, \theta) | a \leq \theta \leq b, a \leq r \leq b\}$ where $0 \leq b - a \leq 2\pi$

$$\iint_R f(x, y) dA = \int_a^b \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta \quad dA = r dr d\theta \quad p: (r, \theta) \mapsto (x, y)$$

alternatively: $\iint_R f(x, y) dA = \int_a^b \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) r dr d\theta \quad \leftarrow n(\theta) \leq r \leq h_2(\theta)\right\}$

surface area (app. of double integrals): S is SURFACE w/ $z = P(x, y)$

$$A(S) = \iint_D \sqrt{P_x^2 + P_y^2 + 1} dA \quad dS = \sqrt{P_x^2 + P_y^2 + 1} dA$$

TRIPLE INTEGRALS: 26 possible orders of integration!

Type 1: $\iiint_R f(x, y, z) dV = \int_a^b \int_{c_1(y)}^{c_2(y)} \int_{d_1(y, z)}^{d_2(y, z)} f(x, y, z) dx dy dz$

Type 2: $\iiint_R f(x, y, z) dV = \int_a^b \int_{c_1(x)}^{c_2(x)} \int_{d_1(x, y)}^{d_2(x, y)} f(x, y, z) dy dx dz$

Type 3: $\iiint_R f(x, y, z) dV = \int_a^b \int_{c_1(z)}^{c_2(z)} \int_{d_1(z, y)}^{d_2(z, y)} f(x, y, z) dx dy dz$

Cylindrical coordinates (polar in SD): (r, θ, z)

$x = r \cos \theta, y = r \sin \theta, z = z$

$r^2 = x^2 + y^2, z = z$

$dV = r dr d\theta dz$

Spherical coordinates (ρ, θ, ϕ)

$x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$

$\rho^2 = x^2 + y^2 + z^2, \phi = \arccos \frac{z}{\rho}$

VECTOR CALCULUS

$\vec{F}(x, y) = \langle P(x, y), Q(x, y) \rangle$ VECTOR FIELD IN $\mathbb{R}^2$: a function F that assigns to each point (x, y) in D a 2D vector $\vec{F}(x, y) = P(x, y)\vec{i} + Q(x, y)\vec{j}$

GRADIENT FIELDS $\vec{F}(x, y) \Rightarrow$ SCALAR FUNCTION

$\nabla f(x, y) = \left\langle \frac{\partial f}{\partial x}(x, y), \frac{\partial f}{\partial y}(x, y) \right\rangle$ GRADIENT VECTOR FIELD ($\mathbb{R}^2$)

$\nabla f(x, y, z) = \left\langle \frac{\partial f}{\partial x}(x, y, z), \frac{\partial f}{\partial y}(x, y, z), \frac{\partial f}{\partial z}(x, y, z) \right\rangle$ GRADIENT VECTOR FIELD ($\mathbb{R}^3$)

vector field $\vec{F}$ is CONSERVATIVE if it is a gradient of some scalar function $\vec{F} = \nabla \phi$

ϕ : POTENTIAL function of $\vec{F}$ $\nabla \phi$ NOT all vector fields conservative

LINE INTEGRALS: single integrals over curves C

line integrals of scalar functions

$I = \int_C f(x, y) ds$ geometric interp: if $f(x, y) \geq 0$, area of curve under f above C

$x = x(t), y = y(t)$ $a \leq t \leq b$ $\Rightarrow ds = \sqrt{x'(t)^2 + y'(t)^2} dt$

Generalizing to 3D: $C: x = x(t), y = y(t), z = z(t)$ $a \leq t \leq b$ $\Rightarrow ds = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} dt$

curve $C: \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j} + z(t)\vec{k}$ $\vec{r}'(t) = x'(t)\vec{i} + y'(t)\vec{j} + z'(t)\vec{k}$ (TANGENT VEC TO C)

$|\vec{r}'(t)| = \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2} \Rightarrow ds = |\vec{r}'(t)| dt$ Arc length: $C: \int_a^b |\vec{r}'(t)| dt$

Thus: $\int_C f(x, y, z) ds = \int_a^b f(x(t), y(t), z(t)) |\vec{r}'(t)| dt$

line integrals of vector fields (vector functions) $\vec{F}$

$\vec{F}(x, y, z) = \langle P(x, y, z), Q(x, y, z), R(x, y, z) \rangle$ $\vec{r}(t) = \langle x(t), y(t), z(t) \rangle$ $\vec{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$

work done by the force field $\vec{F}$ in a moving particle along C : $W = \int_C \vec{F} \cdot d\vec{r}$

$\vec{F} \cdot d\vec{r} = P dx + Q dy + R dz$ line integral of $\vec{F}$ along C Recall: $\vec{r}(t) = (x(t), y(t), z(t))$

$\int_C \vec{F} \cdot d\vec{r} = \int_a^b (P(x(t), y(t), z(t))x'(t) + Q(x(t), y(t), z(t))y'(t) + R(x(t), y(t), z(t))z'(t)) dt$

special cases: $\int_C f(x, y) dx = \int_a^b f(x(t), y(t)) x'(t) dt$ w.r.t. x $\int_C f(x, y) dy = \int_a^b f(x(t), y(t)) y'(t) dt$ w.r.t. y

curve orientation: parametrization of C determines ORIENTATION w/ POSITIVE direction

corresponding to INCREASING values of parameter t

$\int_C f(x, y) dx = - \int_C f(x, y) dx$ $\int_C f(x, y) dy = - \int_C f(x, y) dy$ $\int_C f(x, y) dz = - \int_C f(x, y) dz$

$\Rightarrow \int_C \vec{F} \cdot d\vec{r} = \int_C P dx + Q dy + R dz = - \int_C P dx + Q dy + R dz = - \int_C \vec{F} \cdot d\vec{r}$

fundamental theorem for line integrals

C : smooth curve given by $\vec{r}(t)$ $a \leq t \leq b$

$\oint_C \vec{F} \cdot d\vec{r}$ differentiable func of 2 or 3 variables: $\oint_C \vec{F} \cdot d\vec{r} = \phi(\vec{r}(b)) - \phi(\vec{r}(a))$

$\vec{F}$ is continuous on C

if $\vec{F}$ is conservative: $\oint_C \vec{F} \cdot d\vec{r} = \phi(\vec{r}(b)) - \phi(\vec{r}(a))$ where $\vec{F} = \nabla \phi$

line integral over C values of POTENTIAL at END POINTS

path dependence/independence

alternatively: DEPENDENCE conservative fields $\vec{F}$: INDEPENDENCE

$\oint_C \vec{F} \cdot d\vec{r} \neq 0$ $\oint_C \vec{F} \cdot d\vec{r} = 0$ $\oint_C \vec{F} \cdot d\vec{r} = 0$ $\vec{F} = \nabla \phi$

curve is CLOSED if initial point coincides with terminal point

if $\vec{F}$ path independent $\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = 0$ for closed curve C ($\oint_C \vec{F} \cdot d\vec{r} = 0$)

$\oint_C \vec{F} \cdot d\vec{r}$ independent of path $C \Leftrightarrow \oint_C \vec{F} \cdot d\vec{r} = 0$ for closed paths C

Let D be open and connected and $\vec{F}$ be continuous on D :

$\vec{F}$ conservative $\Leftrightarrow \oint_C \vec{F} \cdot d\vec{r}$ path independent $\Leftrightarrow \oint_C \vec{F} \cdot d\vec{r} = 0$

$\vec{F}$ conservative $\Leftrightarrow P, Q$ have continuous 1st-order derivatives $\Rightarrow \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

simply connected if simple, closed curve in D encloses only points in D

for simply connected region: $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} \Leftrightarrow \vec{F} = \langle P, Q \rangle$ conservative

$\oint_C \vec{F} \cdot d\vec{r} = \int_a^b (\vec{F}(\vec{r}(t)) \cdot \vec{r}'(t)) dt = \int_C \vec{F} \cdot d\vec{r} = \int_C \text{Proj}_{\vec{F}} \vec{F} ds$

CURL and DIVERGENCE

CURL: curl $\vec{F} = \nabla \times \vec{F}$ DIVERGENCE: div $\vec{F} = \nabla \cdot \vec{F}$

2D conservative vector fields have no rotation: curl $\vec{F} = \text{curl}(\vec{F}) = 0$

3D: $\vec{F}$ is scalar or 3 variables w/ continuous 2nd order partials: curl $(\nabla \times \vec{F}) = 0$

$\vec{F}$ conservative $\Rightarrow$ no rotation: curl $\vec{F} = 0$ if domain no holes, iff:

div curl $\vec{F} = \nabla \cdot (\nabla \times \vec{F}) = 0$

if $\vec{F} = P\vec{i} + Q\vec{j} + R\vec{k}$ has continuous 2nd order partials: div curl $\vec{F} = 0$

$\Rightarrow$ curl (grad ϕ) = $\nabla \times (\nabla \phi) = 0$ div (curl $\vec{F}$) = $\nabla \cdot (\nabla \times \vec{F}) = 0$

div (grad ϕ) = $\nabla \cdot (\nabla \phi) = \Delta \phi$ (LAPLACE OPERATOR)

GREEN'S THEOREM: $\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx dy$ (sign determined by orientation)

C is positively oriented, piecewise smooth, simple closed curve

D is region bounded by C , P, Q have continuous partials

Green's theorem in vector notation:

$C: \vec{r}(t) = x(t)\vec{i} + y(t)\vec{j}, d\vec{r} = dx\vec{i} + dy\vec{j}$ $\oint_C \vec{F} \cdot d\vec{r} = \iint_D \text{curl } \vec{F} \cdot \vec{k} dA$

$\vec{F} = P(x, y)\vec{i} + Q(x, y)\vec{j}$ $\oint_C \vec{F} \cdot d\vec{r} = \iint_D (\text{curl } \vec{F} \cdot \vec{k}) dA$

several vector forms of Green's theorem: $\oint_C \vec{F} \cdot d\vec{r} = \iint_D \text{div } \vec{F} dA$

parametric surfaces - surface area

$\vec{r}(u, v) = x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}$ $(u, v) \in D$ $\vec{r}_u = \frac{\partial \vec{r}}{\partial u} = \frac{\partial x}{\partial u}\vec{i} + \frac{\partial y}{\partial u}\vec{j} + \frac{\partial z}{\partial u}\vec{k}$

$\Delta s = \sqrt{r_u^2 + r_v^2} du dv$ $A(S) = \iint_D \Delta s du dv$ $A(S) = \iint_D |\vec{r}_u \times \vec{r}_v| du dv$

SURFACE INTEGRALS

(1) Surface Integrals of SCALAR FUNCTIONS

$I = \iint_S f(x, y, z) dS$ Parametric Surfaces: $\vec{r}(u, v) = x(u, v)\vec{i} + y(u, v)\vec{j} + z(u, v)\vec{k}$ $dS = |\vec{r}_u \times \vec{r}_v| du dv$

Surfaces as Graphs of Func $z = z(x, y)$ $dS = \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy$

(2) Surface Integrals of VECTOR FUNCTIONS

Motivation: $\vec{F} = \rho \vec{v}$, ρ fluid density, $\vec{v}$ fluid velocity, S sphere of radius $R \Rightarrow$ MASS FLUX: $\iint_S \vec{F} \cdot \vec{n} dS$

oriented surface For $\vec{r}(u, v)$: NATURAL ORIENTATION: $\vec{n} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}$ For $z = g(x, y)$: $\vec{n} = \frac{-g_x\vec{i} - g_y\vec{j} + \vec{k}}{\sqrt{1 + g_x^2 + g_y^2}}$

$\vec{n}$ provides ORIENTATION: POSITIVELY ORIENTED: unit normal OUTWARD NEGATIVELY ORIENTED: unit normal INWARD

ORIENTED: IF $\vec{F}$ continuous vector field on oriented surface S w/ unit normal vec. $\vec{n}$

SURFACE: $d\vec{S} = \vec{n} dS$ SURFACE INTEGRAL OF $\vec{F}$ OVER S : $\iint_S \vec{F} \cdot d\vec{S} = \iint_S \vec{F} \cdot \vec{n} dS$

FOR PARAMETRIC SURFACES $d\vec{S} = \begin{cases} \vec{r}_u \times \vec{r}_v du dv & \text{for } \vec{r} = \vec{r}(u, v) \\ \left(\frac{\partial x}{\partial x} \frac{\partial y}{\partial y} - \frac{\partial x}{\partial y} \frac{\partial y}{\partial x} \right) \vec{k} du dv & \text{for } z = z(x, y) \end{cases}$

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